Hasan Bukhari

Software Engineering • Systems • Backend & Applications

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Education

University of Southern Mississippi — **Expected Dec 2027**
B.S. in Computer and Information Sciences; Minor in Biological Sciences — *Hattiesburg, MS*

GPA: 4.00 • President's List • Academic Excellence Scholar

Software Engineering Experience

Data Analyst Intern, Health, Safety & Environment — **Jul 2026 – Aug 2026**
Engro Technical Services — *Karachi, Pakistan*

- Engineered Python, FastAPI, and Power Query pipelines for risk scoring, scheduled reporting, and automated change monitoring across approximately 1,750 controlled engineering and HSE files.
- Built Power BI data models, DAX measures, and executive analytics over 130,000+ incident, near-miss, inspection, audit, waiver, and corrective-action records.

Student Worker, Operations Technology & Event Support — **Jun 2024 – Present**
Thad Cochran Center / Union Operations, University of Southern Mississippi — *Hattiesburg, MS*

- Built SetItUp and UnionInventoryPro to digitize room requests, event layouts, equipment tracking, and recurring inventory audits for university operations.

Undergraduate Researcher, Ichthyology Lab — **Spring 2026 – Present**
University of Southern Mississippi — *Hattiesburg, MS*

- Build reproducible Python workflows and chromosome-correspondence matrices to analyze chromosome-scale genome assemblies and comparative-genomics evidence.

Software Engineering Intern — **Aug 2023 – Feb 2024**
Shahid I.T. Consultants — *Karachi, Pakistan*

- Automated recurring record-update workflows in Python, reducing manual processing by 30%; supported integration testing, deployment documentation, and client application maintenance.

Selected Software Projects

BRACE — 1st Place, VOXO Hatchathon 2026 Bio Track | Python, FastAPI, JavaScript, scikit-learn | GitHub — **2026**

- Co-built BRACE (Biological Reliability Assessment across Cohorts and Experiments), a portable bioinformatics workbench for cohort preparation, cross-dataset validation, shift characterization, and reproducible evidence reporting.
- Engineered dataset acquisition/hashing, API and persistence workflows, release packaging, and 336+ automated tests under a hackathon deadline.

AEGIS — Controlled Document Intelligence Platform | Python, FastAPI, React, TypeScript, Rust/Tauri | GitHub — **2026**

- Built a local-first Windows platform with deterministic file monitoring, version diffing, SHA-256 integrity checks, searchable audit workflows, a React/TypeScript UI, regression tests, migrations, and native packaging.

HEARTLINE — Native Android Lyric Companion | Kotlin, Jetpack Compose, Room, Android Services | GitHub — **2026**

- Built a native Android application with media-session detection, synchronized playback state, offline persistence, resilient service recovery, on-device image rendering, automated tests, lint, and CI builds.

To Ash Again — 2D Narrative Platformer | Python, Pygame, OOP | GitHub — **2025**

- Developed a four-act standalone game over two months with modular player/enemy systems, alternate-character mechanics, inventory and powers, progression, save state, cutscenes, and a multi-phase boss.

WCRE — Windows Checkpoint/Restore Engine | Rust, Win32, Windows PSS, x64 | GitHub — **2026**

- Developing an experimental Windows x64 checkpoint/restart engine with process inspection, PSS-backed checkpoint capture, persistent checkpoint images, restore planning, and controlled virtual-address reconstruction.

Technical Skills

Languages: Python, Rust, C++, Kotlin, Dart, JavaScript/TypeScript, SQL
Applications & Backend: FastAPI, Flask, REST APIs, React, Android, Jetpack Compose, Flutter, Room, Tauri
Systems & Data: Win32 API, Windows PSS, filesystem monitoring, pandas, NumPy, SciPy, scikit-learn, NetworkX, Power BI, DAX, Power Query
Tools & Infrastructure: Git/GitHub, GitHub Actions, Linux/WSL, Cargo, Gradle, PlatformIO, CI/CD, native Windows packaging

Leadership & Honors

Leadership: President & Founder, Student Poets Association and Creative Writing Club, USM.

Selected Honors: 1st Place, VOXO Hatchathon 2026 Bio Track; Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026).